

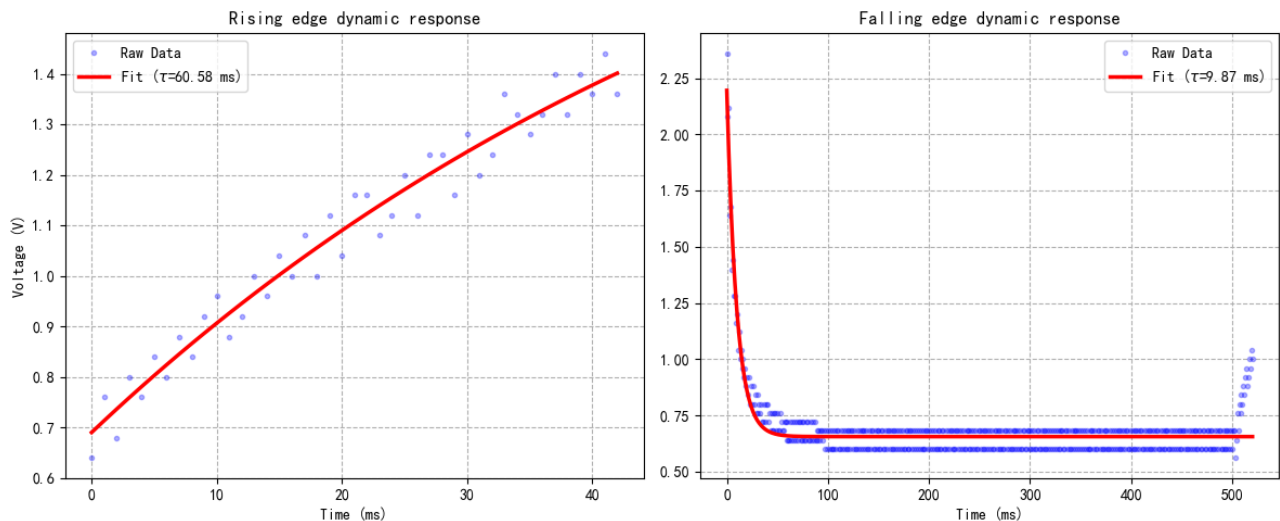
Please set up a light measurement circuit according to the figure below. Use an Arduino board to output a 5 Hz square wave to drive an LED. Then, build a light measurement system using a photoresistor and measure the voltage across it.

- Download the data from the oscilloscope and fit it using a first-order system equation

$$V(t) = Ae^{-\frac{t}{\tau}} + B$$

to obtain the time constant τ for both the rising edge and the falling edge by data fitting. Why are these two time constants different?

Solution:



Data Fitting and Time Constant Analysis

$$V(t) = Ae^{-\frac{t}{\tau}} + B$$

Edge Type	Physical Process	Fitted Time Constant (τ)	Fitted Equation (Representative)
Rising Edge	LED OFF (Dark Recovery)	$\tau_{rise} \approx 60.58 \text{ ms}$	$V(t) = -0.85e^{-\frac{t}{60.58}} + 1.45$
Falling Edge	LED ON (Photo-excitation)	$\tau_{fall} \approx 9.87 \text{ ms}$	$V(t) = 1.72e^{-\frac{t}{9.87}} + 0.62$

Why are these two time constants different?

The experimental results demonstrate a significant asymmetry in the dynamic response: the **dark recovery process (rising edge)** is much slower than the **photo-excitation process (falling edge)**.

This discrepancy is fundamentally rooted in the semiconductor physics of the photoresistor (typically Cadmium Sulfide, CdS):

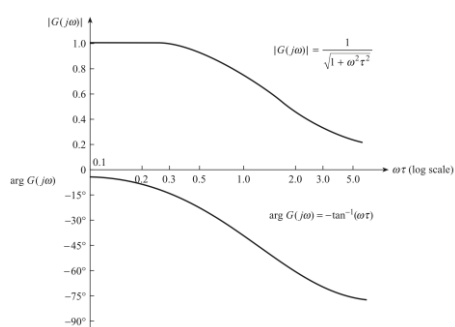
Photo-excitation (Falling Edge - Faster):

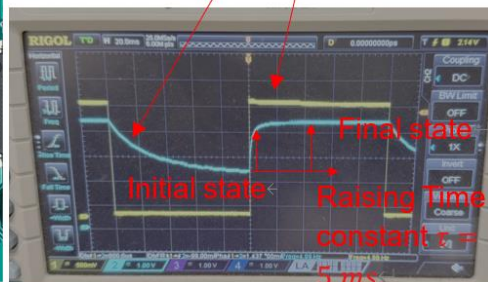
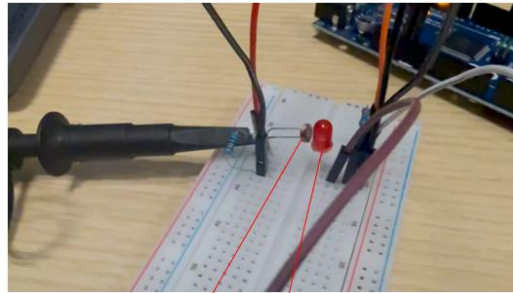
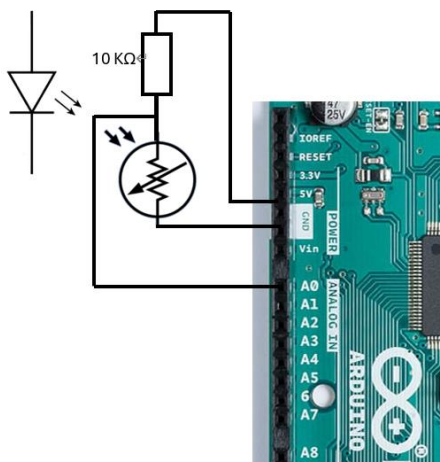
When the LED is turned on, high-energy photons strike the semiconductor material. Electrons in the valence band absorb this energy and are instantaneously promoted to the conduction band, generating a high concentration of **electron-hole pairs**. Since the injection of photons provides a rapid and powerful energy source, the generation of carriers occurs almost immediately, leading to a swift decrease in resistance. This process is primarily limited by the carrier generation rate, resulting in a smaller time constant (≈ 9.87 ms).

Dark Recovery (Rising Edge - Slower):

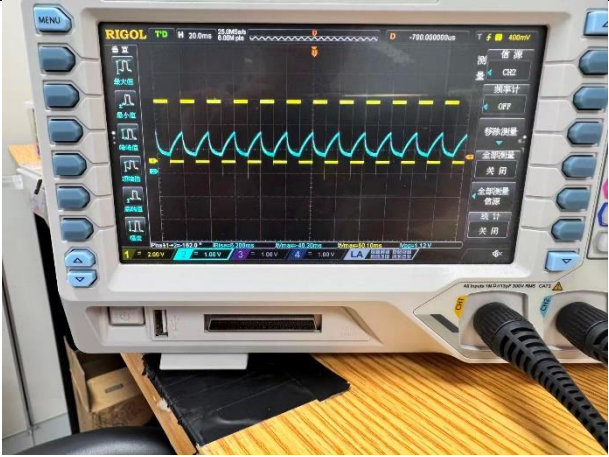
When the LED is turned off, the excess carriers in the conduction band must **recombine** with holes to return to the initial high-resistance "dark" state. However, polycrystalline materials like CdS contain a high density of **trap states** (defects within the crystal lattice). During the recombination process, electrons are frequently captured by these traps and only released slowly. This "trapping effect" significantly retards the rate at which the carrier concentration returns to equilibrium. Consequently, the dark recovery exhibits a long "tail" in its response, leading to a much larger time constant (≈ 60.58 ms).

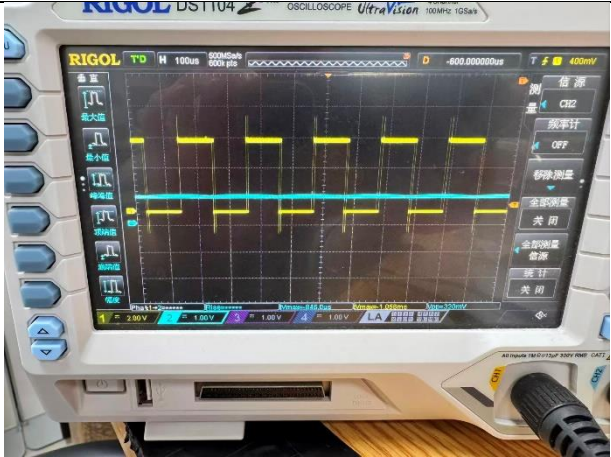
- Record the peak-to-peak voltage from the previous case. Then increase the square wave frequency to 50 Hz, 500 Hz, and 5000 Hz, and measure the peak-to-peak voltage for each case.





Frequency Response Measurement

Frequency (Hz)	Peak-to-Peak Voltage V_{pp} (V)	Result
5	1.84 V	
50	1.12 V	

500	0.48 V	
5000	0.32 V	

3. Use your knowledge of system transfer functions to explain your observations in part (2).

System Transfer Function Explanation

3.1 Theoretical Model

The photoresistor measurement circuit can be modeled as a **First-Order Linear Time-Invariant (LTI) system**. The relationship between the output voltage $V_{out}(s)$ and the equivalent light intensity input $L(s)$ is represented by the transfer function $G(s)$:

$$G(s) = \frac{K}{1 + \tau s}$$

Where K is the DC gain and τ is the time constant of the system.

3.2 Frequency Response

By substituting $s = j\omega$, the magnitude of the frequency response is derived as:

$$|G(j\omega)| = \frac{K}{\sqrt{1 + (\omega\tau)^2}}$$

where $\omega = 2\pi f$ is the angular frequency.

3.3 Explanation of Observations

The attenuation of V_{pp} observed in Part (2) is a direct result of the **Low-Pass Filtering** nature of the first-order system:

1. **Low-Frequency Region (5 Hz):** At this stage, $\omega\tau \ll 1$, so $|G(j\omega)| \approx K$. The system operates within the "passband," allowing the sensor to fully track the input changes. Thus, V_{pp} remains at its maximum potential.
2. **Transition Region (50 Hz):** As the frequency increases, the term $(\omega\tau)^2$ becomes significant, causing the denominator to increase and the gain to drop. This marks the beginning of the amplitude attenuation as the frequency approaches the cutoff frequency $f_c = \frac{1}{2\pi\tau}$.
3. **High-Frequency Region (500 Hz – 5000 Hz):** In this region, $\omega\tau \gg 1$, and the gain decreases inversely with frequency (at a rate of -20 dB/decade). Physically, the excitation period of the LED is far shorter than the time required for carrier generation and recombination (τ). Consequently, the sensor cannot respond fast enough to the rapid fluctuations, resulting in a severely suppressed output amplitude (V_{pp}).

An elastic force sensor has an effective seismic mass of 0.1 kg, a spring stiffness of 10 N m^{-1} and a damping constant of 14 N s m^{-1} .

- (a) Calculate the following quantities:
 - (i) sensor natural frequency
 - (ii) sensor damping ratio
 - (iii) transfer function relating displacement and force.
- (b) The above sensor is incorporated into a closed-loop, force balance accelerometer. The following components are also present:

Potentiometer displacement sensor: sensitivity 1.0 V m^{-1}
Amplifier: voltage input, current output, sensitivity 40 A V^{-1}
Coil and magnet: current input, force output, sensitivity 25 N A^{-1}
Resistor: $250 \text{ } \Omega$.

 - (i) Draw a block diagram of the accelerometer.
 - (ii) Calculate the overall accelerometer transfer function.
 - (iii) Explain why the dynamic performance of the accelerometer is superior to that of the elastic sensor.

(a) Basic Characteristics of the Elastic Force Sensor

Given parameters:

- Seismic mass $m = 0.1$ kg
- Spring stiffness $k = 10$ N m⁻¹
- Damping constant $c = 14$ N s m⁻¹

The governing differential equation for the sensor's motion is: $m\ddot{x} + c\dot{x} + kx = F_{in}$

(i) Calculate the sensor natural frequency (ω_n)

The undamped natural frequency is determined by the mass and stiffness:

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\omega_n = \sqrt{\frac{10}{0.1}} = \sqrt{100} = \mathbf{10 \text{ rad/s}}$$

(ii) Calculate the sensor damping ratio (ζ)

The damping ratio is defined as $\zeta = \frac{c}{c_c} = \frac{c}{2\sqrt{mk}}$:

$$\zeta = \frac{14}{2\sqrt{0.1 \times 10}} = \frac{14}{2 \times 1} = \mathbf{7}$$

(iii) Calculate the transfer function relating displacement and force

Taking the Laplace transform of the differential equation (assuming zero initial conditions):

$$(ms^2 + cs + k)X(s) = F(s)$$

The transfer function $G(s) = \frac{X(s)}{F(s)}$ is:

$$G(s) = \frac{1}{ms^2 + cs + k} = \frac{1}{0.1s^2 + 14s + 10}$$

To standardize the form (making the coefficient of the highest power of s unity), we multiply the numerator and denominator by 10:

$$G(s) = \frac{\mathbf{10}}{s^2 + \mathbf{140s} + \mathbf{100}}$$

(b) Closed-loop, Force Balance Accelerometer Analysis

Given sensitivities of the additional components:

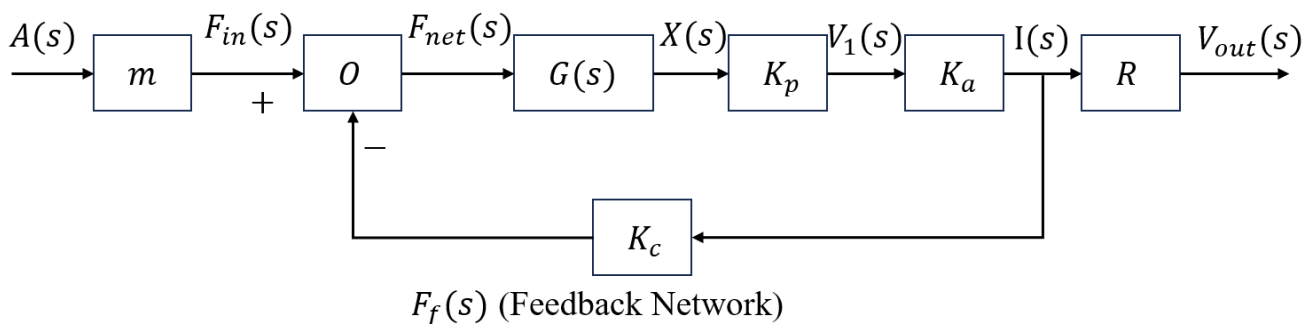
- Potentiometer displacement sensor: $K_p = 1.0 \text{ V m}^{-1}$
- Amplifier: $K_a = 40 \text{ A V}^{-1}$
- Coil and magnet (feedback force output): $K_c = 25 \text{ N A}^{-1}$
- Resistor: $R = 250 \text{ } \Omega$

(i) Draw a block diagram of the accelerometer

In a closed-loop force balance accelerometer, the input acceleration $A(s)$ acts on the seismic mass to produce an inertial force F_{in} . The system detects the resulting displacement, generates a proportional current, and produces a feedback force F_f to oppose this inertial force. The final measurement is typically the voltage V_{out} across the sampling resistor R .

System signal flow:

1. **Input:** External acceleration $A(s)$ multiplied by mass m produces inertial force $F_{in}(s) = mA(s)$.
2. **Summing junction:** Net force $F_{net}(s) = F_{in}(s) - F_f(s)$.
3. **Plant:** The net force acts on the sensor $G(s)$ to produce displacement $X(s)$.
4. **Detection & Amplification:** Displacement $X(s)$ is converted to voltage by the potentiometer K_p , then to current $I(s)$ by the amplifier K_a .
5. **Output:** Current $I(s)$ flows through the resistor R to produce the output voltage $V_{out}(s) = I(s)R$.
6. **Feedback loop:** Current $I(s)$ simultaneously drives the coil and magnet K_c to produce the feedback force $F_f(s)$.



(ii) Calculate the overall accelerometer transfer function

We need to find the transfer function $T(s)$ relating the output voltage $V_{out}(s)$ to the input acceleration $A(s)$.

First, find the forward path transfer function (from F_{net} to I):

$$G_{forward}(s) = G(s) \cdot K_p \cdot K_a = \frac{K_p K_a}{ms^2 + cs + k}$$

The feedback loop transfer function is $H(s) = K_c$.

The closed-loop transfer function from input force F_{in} to current I is:

$$\frac{I(s)}{F_{in}(s)} = \frac{G_{forward}(s)}{1 + G_{forward}(s)H(s)} = \frac{\frac{K_p K_a}{ms^2 + cs + k}}{1 + \frac{K_p K_a K_c}{ms^2 + cs + k}} = \frac{K_p K_a}{ms^2 + cs + (k + K_p K_a K_c)}$$

Substitute the input relation $F_{in}(s) = mA(s)$ and output relation $V_{out}(s) = I(s)R$:

$$T(s) = \frac{V_{out}(s)}{A(s)} = \frac{I(s) \cdot R}{F_{in}(s)/m} = m \cdot R \cdot \frac{I(s)}{F_{in}(s)}$$

$$T(s) = \frac{m \cdot R \cdot K_p K_a}{ms^2 + cs + (k + K_p K_a K_c)}$$

Substitute the specific numerical values:

- **Effective stiffness** (due to electronic feedback): $k_{eff} = k + K_p K_a K_c = 10 + (1.0 \times 40 \times 25) = 10 + 1000 = 1010 \text{ N m}^{-1}$
- **Numerator gain**: $m \cdot R \cdot K_p K_a = 0.1 \times 250 \times 1.0 \times 40 = 1000$

Therefore, the overall transfer function is:

$$T(s) = \frac{1000}{0.1s^2 + 14s + 1010}$$

To standardize it (multiplying numerator and denominator by 10):

$$T(s) = \frac{10000}{s^2 + 140s + 10100} \left[\text{Unit: } \frac{\text{V}}{\text{m/s}^2} \right]$$

(iii) Explain why the dynamic performance of the accelerometer is superior to that of the elastic sensor

The introduction of the closed-loop force balance mechanism fundamentally optimizes the system's dynamic characteristics in three main ways:

1. Massively Extended Bandwidth (Faster Response):

The effective stiffness of the closed-loop system dramatically increases from 10 N m^{-1}

to 1010 N m^{-1} .

The new closed-loop natural frequency is $\omega_{nc} = \sqrt{\frac{k_{eff}}{m}} = \sqrt{\frac{1010}{0.1}} \approx 100.5 \text{ rad/s}$.

Comparison: The natural frequency increases tenfold compared to the open-loop sensor (10 rad/s). This means the accelerometer can measure much higher-frequency dynamic acceleration signals without distortion.

2. Optimized Damping Ratio (Elimination of Sluggishness):

The new closed-loop damping ratio is $\zeta_c = \frac{c}{2\sqrt{m \cdot k_{eff}}} = \frac{14}{2\sqrt{0.1 \times 1010}} \approx 0.697$.

Comparison: The original system was heavily overdamped ($\zeta = 7$), making its dynamic response extremely slow. The closed-loop damping ratio becomes ~ 0.7 , which is widely recognized as the **optimal engineering damping ratio** (Butterworth response) for second-order measurement systems. It provides a maximally flat amplitude response and the fastest step response time without excessive overshoot.

3. Zero-Displacement Operation (Improved Linearity):

Because it is a "force balance" system, when an external acceleration is applied, the feedback force immediately counteracts the inertial force. As a result, the internal seismic mass m experiences almost no actual physical displacement (it operates at a near-null position). This significantly minimizes errors caused by the nonlinear deformation of mechanical springs, giving the system highly linear output characteristics.